

# PIPE-Cypher: Automatic Enterprise Benchmark Generation for Text-to-Cypher Systems

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## Abstract

Enterprises increasingly use property graphs and Cypher to analyze fraud, risk, identity, customer, and operational data. Public Text2Cypher benchmarks rarely reflect private schemas, terminology, access patterns, or difficulty distributions. We present PIPE-Cypher, a local-model pipeline for generating organization-specific natural-language-to-Cypher benchmarks. PIPE-Cypher combines schema introspection, reverse-query grounding, constrained Cypher generation, deterministic read-only and schema validation, execution feedback, repair, diversity controls, and LLM-judge review. The system is designed for industry settings where benchmark generation must avoid paid APIs, preserve data governance, and remain repeatable as graphs evolve.

## 1 Introduction

Property graphs encode enterprise relationships directly: account transfers, identity entitlements, access paths, customer interactions, and fraud rings. Cypher is a common query language for these graphs. As LLMs become natural-language interfaces for graph analytics, organizations need reliable benchmarks for natural-language-to-Cypher systems on their own schemas.

Static public benchmarks are insufficient for this setting. They cannot contain private labels, relationship types, categorical values, or domain-specific wording, and they do not track schema evolution. PIPE-Cypher addresses this gap by generating private, repeatable, execution-grounded NL-Cypher benchmarks.

## 2 Related Work

Recent Text2Cypher resources, including Mind the Query [2], SyntheT2C [19], Auto-Cypher [15], and Text2Cypher [10], show that high-quality Cypher data requires more than asking an LLM for query pairs. These works motivate schema grounding, execution checks, synthetic generation, verification, and complexity-aware evaluation. PIPE-Cypher differs by targeting private enterprise benchmark generation as the primary artifact: organizations bring their own graph, run local models, enforce Cypher constraints, and receive an executable benchmark with diversity and judge metadata.

Text-to-SQL benchmarks such as Spider 2.0 [5] and BIRD [6] motivate realistic database tasks, execution-based evaluation, and enterprise workflow complexity. CIKM AutoQuery [18] motivates strategy-level workload generation analysis and separating workload quality from model quality. LDBC FinBench [13] and SNB [12] provide graph workloads suitable for industry-oriented evaluation.

For generated benchmark quality, PIPE-Cypher combines text-generation diversity diagnostics with text-to-query structure metrics. Distinct-n [7] and self-BLEU-style redundancy checks [20] capture lexical repetition, while schema coverage, query-signature diversity, structural feature rates, and normalized entropy over graph/category/difficulty cells capture properties that matter for executable Cypher benchmarks.

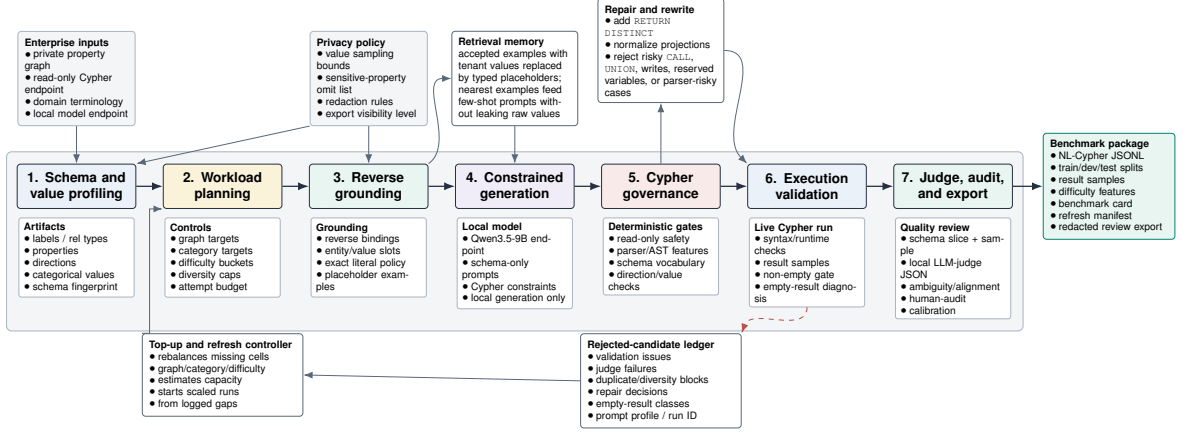


Figure 1: PIPE-Cypher benchmark generation pipeline. The key industry additions beyond static Text2Cypher dataset construction are privacy/value policies, reverse grounding, Cypher governance, execution diagnostics, local judge calibration, and benchmark export/refresh.

| Gate                              | Evidence checked                                           | Failure mode caught                       |
|-----------------------------------|------------------------------------------------------------|-------------------------------------------|
| Read-only safety                  | blocked Cypher write/admin tokens                          | destructive or operational queries        |
| Schema validity                   | labels, relationship types, properties, categorical values | hallucinated graph vocabulary or values   |
| Direction validity                | observed relationship directions                           | reversed or invalid traversals            |
| Cypher analysis and normalization | structure extraction, rewrite safety, RETURN DISTINCT      | unsafe rewrites or duplicate/noisy rows   |
| Question constraints              | quoted values use exact literals                           | fuzzy matching of exact user values       |
| Execution                         | query runs and returns rows                                | syntactic validity without answerability  |
| LLM judge                         | ambiguity, semantic alignment, schema use                  | executable but semantically weak examples |

Table 1: PIPE-Cypher candidate acceptance gates.

### 3 Method

PIPE-Cypher has six stages: schema profiling, template generation, reverse Cypher grounding, constrained Cypher generation and repair, deterministic validation and execution, and LLM-judge review. Deterministic checks enforce read-only safety, syntax shape, schema-visible labels and properties, explicit relationship types, observed relationship directions, schema-provided categorical values, and non-empty execution where required. A lightweight Cypher analyzer extracts return aliases, variables, labels, relationship observations, risky constructs, and rewrite skip reasons so normalization is auditable rather than a silent string edit. The direction gate interprets both outgoing and incoming Cypher arrow syntax before schema checking, and rejects undirected relationship patterns so directionality is an executable constraint rather than only a prompt instruction.

### 4 Implementation

The implementation is a Python package with Neo4j as the experimental backend. The paper frames the contribution around Cypher and property graphs rather than a single vendor. Local models are served through a vLLM/OpenAI-compatible endpoint on ds-serv6, using Qwen3.5-9B for generation and judging.

| Graph    | Target/category | Binding limit | Min seed capacity | All categories meet target |
|----------|-----------------|---------------|-------------------|----------------------------|
| FinBench | 250             | 300           | 300               | Yes                        |
| SNB      | 125             | 200           | 200               | Yes                        |

Table 2: Built-in seed-template capacity after removing the reverse-binding 10-row cap and adding slotted negation/ranking seeds. Capacity is a theoretical scale-readiness check; final examples still must pass validation, execution, diversity, and judge gates.

All reported generation, judging, and downstream evaluation runs use a local Qwen3.5-9B endpoint. This keeps the study aligned with an enterprise deployment pattern in which the benchmark pipeline runs inside the organization’s compute boundary without paid generation APIs.

The built-in FinBench profile is grounded in the public datagen snapshot export and includes typed node and relationship properties. Live schema inspection also records low-cardinality string values as categorical constraints for prompting and validation. The generated Neo4j import script uses uniqueness constraints for nodes and creates, rather than merges, transaction relationships so repeated account-to-account events are preserved. The SNB reference profile is grounded in the official Neo4j/Cypher headers and read-query files.

For generation, PIPE-Cypher uses seeded workload templates with deterministic reverse-binding queries and a mixed mode that prepends proven workload seeds to LLM-proposed templates. Every run records accepted and rejected candidates for yield analysis. Retrieved few-shot examples replace graph-specific values with typed placeholders, preserving query structure while reducing tenant-value leakage and memorized entity reuse. A dependency-light value grounder adds typed prompt annotations for categorical values and reverse-bound entities, covering punctuation variants, possessives, plurals, synonyms, name partials, and small typos.

Inspired by Mind the Query’s prompt-setting analysis, the implementation exposes prompt profiles for schema-only, instructions-only, examples-only, examples-plus-instructions, and full governed PIPE-Cypher generation. These profiles are configured as ablation variants and must pass the same target-size and paper-readiness standards before any result is reported.

For LLM-judge review, the implementation slices schema context to the labels, relationship types, and properties used by the candidate query. This preserves validation against the full schema while keeping local 9B prompts within context limits on larger schemas.

The deterministic Cypher layer uses production-derived constraints: schema-only prompting, exact matching, relationship direction discipline, `RETURN DISTINCT`, reserved variable rejection, categorical values, required contextual return columns, fuzzy value annotations, placeholderized retrieval examples, and parser-aware rewrite boundaries. PIPE-Cypher records parser-style structure features and skips rewrites for risky constructs such as `UNION`, `CALL`, `UNWIND`, `WHERE EXISTS`, multiple `WHERE` clauses, or reserved variables.

We also estimate seed-template capacity before full generation. This check caught and fixed a scale blocker: reverse-binding execution was hard-capped to 10 rows, and no-slot negation/ranking seeds could not support full category targets. PIPE-Cypher now uses the configured binding limit and includes slotted negation and ranking seeds for both graphs.

For arbitrary enterprise onboarding, PIPE-Cypher includes a deployment template for a company’s own graph, read-only credentials, local model endpoint, schema introspection, privacy policy, dry run, scaled run, audit, and redacted export. Configurable value-sampling policies bound which low-cardinality graph values enter schema prompts, and redacted exports replace quoted literals, entity values, and string-valued result samples with stable placeholders for broader internal review.

| Setting                | Value                               |
|------------------------|-------------------------------------|
| Accepted examples      | 3,000                               |
| Primary graph          | LDBC FinBench, 2,000 examples       |
| Secondary graph        | LDBC SNB, 1,000 examples            |
| Categories             | 8 balanced categories, 375 each     |
| Generation/judge model | Local Qwen3.5-9B                    |
| Execution backend      | Neo4j Community, two live databases |
| Judge audit packet     | 80 sampled accepted/rejected pairs  |

Table 3: Full live experimental setup used for the generated benchmark artifact.

| Graph    | Candidates | Accepted | Acceptance | Categories at target |
|----------|------------|----------|------------|----------------------|
| FinBench | 3,404      | 2,000    | 0.588      | 8/8                  |
| SNB      | 1,373      | 1,000    | 0.728      | 8/8                  |
| Total    | 4,777      | 3,000    | 0.628      | 16/16                |

Table 4: Full live generation with local Qwen3.5-9B. Candidate counts include the initial sequential run and category-specific recovery top-ups.

## 5 Experiments

The generated benchmark contains 3,000 accepted examples: 2,000 from LDBC FinBench and 1,000 from LDBC SNB. Categories include simple retrieval, complex retrieval, simple aggregation, complex aggregation, boolean existence, negation/difference, path/temporal transaction, and ranking/top-k. Appendix Table 8 reports graph statistics and the current ICIJ onboarding status.

Downstream Text2Cypher evaluation uses execution accuracy and answer-set precision/recall/F1 as primary correctness metrics. The benchmark evaluator also supports supplementary reference-based text metrics over serialized answer sets and query strings, including ROUGE, BLEU, METEOR, BERTScore, FrugalScore, cosine similarity, Jaro-Winkler similarity, and exact match; Appendix Table 20 lists the supported metrics and citations. We treat these as debugging and near-match diagnostics rather than substitutes for executable correctness.

## 6 Results

The full live run produced 3,000 accepted examples from 4,777 candidates using local Qwen3.5-9B for generation and judging. Category-specific recovery top-ups filled the only under-target categories from the initial sequential run. Every exported example passed read-only, syntax, schema, execution, non-empty result, and judge gates.

The accepted full-run records were exported into a benchmark package with stable example identifiers, train/dev/test splits, result samples, gate metadata, aggregate statistics, and a manifest hash for artifact track-

| Export artifact     | Examples | FinBench | SNB   | Train | Dev | Test |
|---------------------|----------|----------|-------|-------|-----|------|
| Live full benchmark | 3,000    | 2,000    | 1,000 | 2,408 | 296 | 296  |

Table 5: Accepted live full benchmark package with stable IDs, gate metadata, result samples, statistics, and manifest hash 8bc79a53a06b291a.

| Artifact property                   | Value                     |
|-------------------------------------|---------------------------|
| Categories                          | 8 balanced categories     |
| FinBench/category                   | 250                       |
| SNB/category                        | 125                       |
| Difficulty split                    | 1,521 easy / 1,479 medium |
| Unique labels / rel. types          | 7 / 9                     |
| Read/syntax/schema/exec/judge gates | 3,000/3,000               |

Table 6: Distribution and gate summary for the exported full benchmark artifact.

ing. The export contains 3,000 accepted examples: 2,000 FinBench, 1,000 SNB, and 375 accepted examples in every planned category.

We report diversity as a diagnostic, not only as a design target. The full export has perfect category balance and near-perfect difficulty balance, uses 1,115 unique grounded entity values, and exactly quotes grounded values in 82.6% of examples with entity bindings. The reported local-model run still has low query-signature diversity because seeded graph-grounded templates are doing substantial work. This is useful evidence for industry deployment: the artifact is balanced and executable, while the appendix makes template and value concentration visible rather than hiding it.

The full-run failure taxonomy shows where generation effort was spent before export: 1,335 rejected candidates came from diversity or duplicate controls during recovery, 374 from empty execution results, 66 from judge semantic rejection, and only 2 from schema invalidity. This supports the claim that deterministic schema gates made invalid Cypher rare, while refresh-scale generation mostly needs better value/template exploration.

For judge calibration, the released artifacts include an 80-row audit packet sampled from accepted and rejected full-run candidates, with a 40/40 judge accept/reject split and coverage over both graphs and all eight categories. A completed human audit shows 80.0% agreement, Cohen’s  $\kappa = 0.60$ , judge precision and specificity of 1.00, recall of 0.714, and no false accepts in the sample. The judge is therefore conservative: it protects accepted-example quality while rejecting some human-approved candidates and reducing yield. Human labels are used only for post-hoc calibration, not as a generation gate.

Finally, we ran a downstream Text2Cypher evaluation using local Qwen3.5-9B on the exported full test split. The model saw schema text and the natural-language question, generated Cypher, and was evaluated by live execution against the corresponding FinBench or SNB database. The result verifies that the exported artifact can drive downstream model evaluation and gives a difficulty signal: the zero-shot 9B baseline solves simple aggregation examples but struggles on more operational categories.

| Split          | Examples | Parse | Schema | Exec. success | Exec. acc. | Answer F1 |
|----------------|----------|-------|--------|---------------|------------|-----------|
| Live full test | 296      | 0.959 | 0.905  | 0.622         | 0.189      | 0.189     |

Table 7: Downstream Text2Cypher evaluation for local Qwen3.5-9B on the exported full benchmark test split.

## 7 Industry Use

PIPE-Cypher is intended to be rerun when an enterprise graph changes. The generated artifact records the schema snapshot, graph profile, model identifier, validation gates, execution samples, judge scores, difficulty features, and source run for every example. Long-running jobs can be monitored from JSONL records and recovered with category-specific top-up runs that reject questions accepted in earlier runs. This design supports private benchmark refreshes without exposing schemas or values to paid APIs, while still leaving audit hooks for data owners to sample accepted and rejected examples.

The deployment path is designed for arbitrary enterprise schemas: teams provide read-only graph credentials, introspect labels/relationships/properties, choose a bounded categorical-value sampling policy, connect a local model endpoint, run a dry pass, scale generation, calibrate the judge, and export either raw internal artifacts or redacted review artifacts. This makes the FinBench/SNB study a reproducible evaluation setting rather than a hard-coded graph assumption.

## 8 Conclusion

PIPE-Cypher provides a practical path for enterprises to create private, executable, balanced NL-to-Cypher benchmarks. The pipeline combines schema introspection, constrained generation, deterministic validation, execution feedback, diversity control, and automated judge review.

## Limitations

Execution validity does not guarantee semantic correctness. The completed audit suggests a conservative judge with no false accepts in the labeled sample, but larger audits may reveal additional failure modes. Generated artifacts may contain private schema details or sampled values, so organizations should apply privacy redaction and normal data governance controls before broad sharing.

## Ethics Statement

PIPE-Cypher is designed for private benchmark generation under local-model deployment constraints. Organizations using it should restrict benchmark artifacts to authorized users, review sampled values for sensitive content, and document whether judge calibration relied on human annotators.

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| Graph               | Nodes  | Relationships | Labels | Rel. types | Paper status    |
|---------------------|--------|---------------|--------|------------|-----------------|
| FinBench            | 10,006 | 57,622        | 5      | 9          | reported        |
| SNB                 | 34,735 | 70,842        | 14     | 15         | reported        |
| ICIJ Offshore Leaks | –      | –             | 5      | 14         | onboarding only |

Table 8: Study graph statistics. ICIJ is listed as an onboarding workload only until its live generation run passes the same audit standard as FinBench and SNB.

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| Metric                                  | Value                 |
|-----------------------------------------|-----------------------|
| Question Distinct-1                     | 0.041                 |
| Question Distinct-2                     | 0.088                 |
| Question self-BLEU-2 (sampled)          | 0.826                 |
| Unique query-signature ratio            | 0.010                 |
| Top query-signature share               | 0.083                 |
| Category normalized entropy             | 1.000                 |
| Graph-category normalized entropy       | 0.980                 |
| Difficulty normalized entropy           | 1.000                 |
| Label coverage                          | 0.412                 |
| Relationship-type coverage              | 0.375                 |
| Property-name coverage                  | 0.407                 |
| Unique grounded-value ratio             | 0.373                 |
| Grounded values exactly quoted          | 0.826                 |
| Aggregation / negation / ordering rates | 0.500 / 0.125 / 0.125 |

Table 9: Diversity diagnostics for the full exported benchmark. Distinct-n follows text-generation usage; self-BLEU is lower when questions are less redundant; grounded-value metrics are aggregate-only and do not list raw values.

## A Additional Results

### B Claim–Evidence Map

This section maps the main paper claims to the strongest current evidence and to the remaining risks. This is intended as a reviewer-facing audit surface: claims with pending evidence remain marked as such rather than being folded into the main results.

1. **Claim 1.** PIPE-Cypher can generate a private, executable NL-to-Cypher benchmark over enterprise-style property graphs using local models.

*Evidence.* The full local Qwen3.5-9B run exported 3,000 accepted examples: 2,000 FinBench and 1,000 SNB, with 375 examples in every planned workload category and all accepted examples passing read-only, syntax, schema, execution, non-empty-result, and judge gates.

*Key artifacts.* benchmark export; manifest snapshot; paper table: benchmark export; paper table: full results

*Status.* Supported by live local-model evidence.

*Risk.* Generation quality may change with private enterprise schemas beyond FinBench/SNB.

| Failure bucket              | Count | Share of rejected |
|-----------------------------|-------|-------------------|
| Diversity/duplicate control | 1,335 | 0.751             |
| Empty result                | 374   | 0.210             |
| Judge semantic reject       | 66    | 0.037             |
| Schema invalid              | 2     | 0.001             |

Table 10: Failure taxonomy over full-run generation candidates before benchmark export. Accepted examples are excluded from the bucket shares.

| PIPE-Cypher category      | Closest MTQ category | Added enterprise distinction                                 |
|---------------------------|----------------------|--------------------------------------------------------------|
| Simple retrieval          | SR                   | Direct node/edge lookup with exact filters.                  |
| Complex retrieval         | CR                   | Multi-hop or multi-pattern retrieval.                        |
| Simple aggregation        | SA                   | Single aggregation such as count/min/max/average.            |
| Complex aggregation       | CA                   | Grouped or multi-stage aggregation over graph neighborhoods. |
| Boolean existence         | EQ                   | Precise yes/no or existence answer.                          |
| Negation/difference       | CR                   | Absence, anti-join, or difference query.                     |
| Path/temporal transaction | CR/CA                | Temporal or path-oriented transaction neighborhood.          |
| Ranking/top-k             | SA/CA                | Ordered top-k query with explicit limit.                     |

Table 11: Category crosswalk to Mind the Query. PIPE-Cypher keeps the familiar retrieval/aggregation/evaluation-query structure while adding enterprise workloads such as negation, temporal paths, and ranking.

2. **Claim 2.** Cypher-specific governance makes generated benchmark candidates safer and more auditable than prompt-only generation.

*Evidence.* The pipeline validates read-only safety, schema-visible labels, node and relationship properties, relationships, observed relationship direction, categorical values, contextual return columns, parser-style structure, execution success, and LLM-judge review. In the full run, only 2 of 4,777 candidates were schema-invalid after deterministic governance, and all 3,000 exported examples passed the deterministic and judge gates.

*Key artifacts.* code: validator.py; code: cypher\_parser.py; code: question\_constraints.py; snapshot: failure taxonomy; +1 more

*Status.* Supported by code, tests, and full-run failure taxonomy.

*Risk.* Semantic correctness still depends on execution evidence and judge review; the completed 80-row audit is a calibration sample rather than an exhaustive proof.

3. **Claim 3.** Reverse grounding and structured workload seeds are designed to improve reliability under local-model constraints.

*Evidence.* The target-50 FinBench/SNB ablation suite completed 14/14 graph/variant cells, reached every category target for all non-empty/non-unconstrained runs, and passed the paper-readiness audit with logs, model IDs, code revision, run summaries, and gate-rate availability. Appendix-only tables plus yield and gate-rate figures report the audited target-50 evidence while the target-100 follow-up

| Gate or ledger             | Count | Denominator |
|----------------------------|-------|-------------|
| Export accepted            | 3,000 | 3,000       |
| Read-only safety           | 3,000 | 3,000       |
| Syntax validity            | 3,000 | 3,000       |
| Schema/value validity      | 3,000 | 3,000       |
| Execution success          | 3,000 | 3,000       |
| LLM judge pass             | 3,000 | 3,000       |
| Rejected candidates logged | 1,777 | 4,777       |

Table 12: PIPE-Cypher validation cascade for the full export and its logged candidate ledger. Unlike Mind the Query, human review is calibration-only.

| Profile                 | Added constraint                                                | Intended evidence                                     |
|-------------------------|-----------------------------------------------------------------|-------------------------------------------------------|
| Schema-only             | Use only visible schema.                                        | Baseline for schema-grounded prompting.               |
| Instructions            | Exact values, datatype rules, no nested aggregations.           | Targets MTQ-style prompt refinements.                 |
| Examples                | Placeholderized retrieved NL-Cypher pairs.                      | Tests whether examples help without extra governance. |
| Examples + instructions | Few-shot plus explicit rules.                                   | Closest controlled analogue to MTQ Table 5.           |
| Full governed           | Production-derived Cypher hints, AST-safe rewrites, judge gate. | PIPE-Cypher production setting.                       |

Table 13: Prompt-profile plan inspired by Mind the Query’s prompt-variant analysis. Only completed, audited target-50-or-larger results should be promoted into results tables.

and seeded target-50 repeat continue for stronger final reliability and variance claims.

*Key artifacts.* script: run live ablation suite; script: monitor remote ablation suite; script: monitor remote ablation queue; script: collect remote ablation suite; +9 more

*Status.* Supported by audited target-50 appendix evidence; target-100 and seeded target-50 repeat runs are still pending for stronger final reliability claims.

*Risk.* A single target-50 suite is the minimum publishable ablation scale, not the strongest possible evidence. If target-100 or repeated-seed runs complete, final claims should use those sensitivity results instead of relying only on target-50.

- Claim 4.** The benchmark controls category and difficulty balance while exposing residual diversity limits.

*Evidence.* The full export has perfect category balance, planned 2:1 FinBench/SNB graph mix, and a near-even easy/medium split. Diversity diagnostics report Distinct-n, sampled self-BLEU-2, query-signature diversity, normalized entropy, schema coverage, structural feature rates, and aggregate value-grounding diagnostics. The full export uses 1,115 unique grounded entity values and exactly quotes grounded values in 82.6% of examples with entity bindings, making template and value concentration visible instead of hidden.

| Dimension              | Mind the Query              | PIPE-Cypher                               |
|------------------------|-----------------------------|-------------------------------------------|
| Generation review gate | Manual logical review       | Deterministic gates + local LLM judge     |
| Human effort           | Reported 1,400 person-hours | 80-row post-hoc judge calibration audit   |
| Private values         | Public benchmark values     | Configurable sampling and redacted export |
| Refresh                | Static dataset release      | Rerunnable private benchmark factory      |
| Model endpoint         | Gemini for generation       | Local Qwen3.5-9B endpoint                 |

Table 14: Industry deployment contrast with Mind the Query. PIPE-Cypher focuses on private refreshable benchmark generation rather than a one-time public dataset.

| Setting           | Graph    | Records | Accepted | Acceptance | Categories at target |
|-------------------|----------|---------|----------|------------|----------------------|
| Unconstrained LLM | FinBench | 0       | 0        | 0.000      | 0/8                  |
| Unconstrained LLM | SNB      | 0       | 0        | 0.000      | 0/8                  |
| Reverse-only      | FinBench | 400     | 400      | 1.000      | 8/8                  |
| Reverse-only      | SNB      | 402     | 400      | 0.995      | 8/8                  |
| Validators+repair | FinBench | 400     | 400      | 1.000      | 8/8                  |
| Validators+repair | SNB      | 402     | 400      | 0.995      | 8/8                  |
| No retrieval      | FinBench | 429     | 400      | 0.932      | 8/8                  |
| No retrieval      | SNB      | 402     | 400      | 0.995      | 8/8                  |
| No rewrite        | FinBench | 423     | 400      | 0.946      | 8/8                  |
| No rewrite        | SNB      | 402     | 400      | 0.995      | 8/8                  |
| No LLM judge      | FinBench | 401     | 400      | 0.998      | 8/8                  |
| No LLM judge      | SNB      | 406     | 400      | 0.985      | 8/8                  |
| Full PIPE-Cypher  | FinBench | 416     | 400      | 0.962      | 8/8                  |
| Full PIPE-Cypher  | SNB      | 402     | 400      | 0.995      | 8/8                  |

Table 15: Live target-50 ablation evidence with local Qwen3.5-9B. Each graph run targets 50 accepted examples per category.

*Key artifacts.* snapshot: diversity report; paper table: diversity; paper figure: diversity diagnostics; paper figure: full export distribution

*Status.* Supported by full-export statistics and diversity diagnostics.

*Risk.* Low query-signature diversity in the reported local-model run remains a limitation and ablation target.

5. **Claim 5.** The generated benchmark is useful for downstream Text2Cypher evaluation.

*Evidence.* A zero-shot local Qwen3.5-9B downstream evaluation over the 296-example full test split reports parse validity, schema validity, execution success, execution accuracy, answer F1, per-category breakdowns, fixed-seed bootstrap uncertainty intervals, and a row-level error taxonomy. The model reaches 0.189 execution accuracy with a 95% bootstrap interval of [0.145, 0.233] and 0.622 execution success with interval [0.564, 0.676]. The error taxonomy classifies 240 incorrect rows into answer mismatches, execution failures, schema invalid predictions, and parse-invalid predictions, showing the benchmark can discriminate easy aggregation from harder operational categories and can explain model failure modes.

| Setting           | Graph    | Read-only | Syntax | Schema | Exec. | Judge |
|-------------------|----------|-----------|--------|--------|-------|-------|
| Unconstrained LLM | FinBench | 0.000     | 0.000  | 0.000  | 0.000 | 0.000 |
| Unconstrained LLM | SNB      | 0.000     | 0.000  | 0.000  | 0.000 | 0.000 |
| Reverse-only      | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 1.000 |
| Reverse-only      | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.995 |
| Validators+repair | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 1.000 |
| Validators+repair | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.995 |
| No retrieval      | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.932 |
| No retrieval      | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.995 |
| No rewrite        | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.946 |
| No rewrite        | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.995 |
| No LLM judge      | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.998 |
| No LLM judge      | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.985 |
| Full PIPE-Cypher  | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.962 |
| Full PIPE-Cypher  | SNB      | 1.000     | 1.000  | 0.995  | 0.995 | 0.995 |

Table 16: Quality-gate rates for the live target-50 ablation suite. Rates are computed over all generated records in each graph/setting.

*Key artifacts.* downstream summary; snapshot: downstream uncertainty; snapshot: downstream error report; research note: downstream evaluation; +6 more

*Status.* Supported by live downstream evaluation against FinBench/SNB databases, with bootstrap uncertainty computed from row-level outputs.

*Risk.* Only one downstream model has been evaluated so far.

- Claim 6.** Automated LLM-judge review can replace human-in-the-loop generation gates while still exposing judge risk.

*Evidence.* The pipeline uses deterministic validation plus LLM-judge review as the generation gate. A post-hoc 80-row judge audit packet preserves 40 judge accepts, 40 judge rejects, both graphs, and all eight categories. The completed human audit reports 80.0% agreement, Cohen’s kappa 0.60, balanced accuracy 0.857, judge precision 1.00, judge specificity 1.00, judge recall 0.714, zero false accepts, and 16 false rejects. The analyzer requires complete labels for paper-facing calibration, and the tracked summary avoids committing raw value-bearing audit rows.

*Key artifacts.* code: judge.py; code: calibration.py; code: judge\_audit\_packet.py; script: create judge annotation sheets; +6 more

*Status.* Supported by completed human audit summary.

*Risk.* The judge appears conservative on this sample; larger audits or different enterprise schemas may reveal false accepts.

- Claim 7.** PIPE-Cypher supports arbitrary enterprise schema onboarding with configurable privacy redaction and value-sampling policies.

*Evidence.* The repo includes an enterprise deployment guide, an enterprise template config, privacy config fields, schema introspection that reads categorical-value thresholds, maximum sampled value

| Audit packet property         | Value         |
|-------------------------------|---------------|
| Rows                          | 80            |
| Judge accept / reject         | 40 / 40       |
| FinBench / SNB rows           | 48 / 32       |
| Easy / medium rows            | 26 / 54       |
| Labeled rows                  | 80            |
| Agreement / Cohen’s $\kappa$  | 0.800 / 0.600 |
| Judge precision / recall      | 1.000 / 0.714 |
| Judge specificity             | 1.000         |
| False accepts / false rejects | 0 / 16        |

Table 17: Post-hoc judge calibration from the completed 80-row human audit. Labels are used only to calibrate the automated judge, not as a generation gate.

length, and omitted-property patterns from config, deterministic redaction helpers, and a redacted benchmark export CLI. The redaction policy replaces quoted Cypher literals, question mentions, entity values, and string-valued result samples with stable placeholders while preserving query structure and gate metadata. The repo now also includes an ICIJ Offshore Leaks onboarding profile, config, download/load helpers, and graph plan for a public finance/compliance property graph beyond the LDBC study workloads.

*Key artifacts.* research note: enterprise deployment guide; research note: icij offshoreleaks graph plan; enterprise\_template.yaml; icij\_offshoreleaks\_smoke.yaml; +10 more

*Status.* Supported by docs, config, code, deterministic tests, and a concrete third-graph onboarding plan; live ICIJ generation remains pending.

*Risk.* ICIJ is public rather than private/anonymized enterprise data. The strongest industry validation would still come from a real tenant graph or a completed audited ICIJ live run.

## C Prompt Contracts

PIPE-Cypher treats prompts as versioned implementation artifacts. The list summarizes the prompt contracts used for generation, repair, judging, and downstream evaluation; hashes fingerprint the full prompt constants in the codebase.

| Metric             | Point | 95% CI         | SE    | N   |
|--------------------|-------|----------------|-------|-----|
| Parse valid        | 0.959 | [0.936, 0.980] | 0.012 | 296 |
| Schema valid       | 0.905 | [0.872, 0.939] | 0.017 | 296 |
| Execution success  | 0.622 | [0.564, 0.676] | 0.028 | 296 |
| Execution accuracy | 0.189 | [0.145, 0.233] | 0.022 | 296 |
| Answer F1          | 0.189 | [0.149, 0.236] | 0.023 | 296 |

Table 18: Bootstrap uncertainty for downstream Text2Cypher evaluation on the full exported test split. Intervals use row-level resampling with a fixed seed and are intended for appendix reporting.

| Downstream outcome | Count | Share of incorrect |
|--------------------|-------|--------------------|
| Answer mismatch    | 128   | 0.533              |
| Execution failed   | 72    | 0.300              |
| Schema invalid     | 28    | 0.117              |
| Parse invalid      | 12    | 0.050              |

Table 19: Downstream Text2Cypher failure taxonomy for local Qwen3.5-9B on the full exported test split. Shares exclude exact-answer matches.

1. **Template generation.** Stage: Workload proposal. SHA-256: 10b25b.  
*Contract.* Schema-only labels, relationships, properties, and categorical values; Realistic enterprise analyst wording; At most two typed slots and JSON-only output
2. **Reverse binding.** Stage: Graph grounding. SHA-256: 48e949.  
*Contract.* Read-only MATCH/WHERE/RETURN DISTINCT/LIMIT only; Slot variables named exactly as requested; Forward relationship directions from the schema
3. **Cypher generation.** Stage: Candidate query. SHA-256: 61c557.  
*Contract.* Only schema-visible constructs and observed directions; RETURN DISTINCT for set returns and exact equality for quoted values; Context columns, categorical hints, placeholderized retrieval, and no writes
4. **Repair.** Stage: Validation feedback. SHA-256: 56c419.  
*Contract.* Preserve question intent while fixing validation or execution issues; Keep query read-only and schema-grounded; Return only corrected Cypher
5. **LLM judge.** Stage: Quality gate. SHA-256: 421c7b.

| Metric       | PIPE-Cypher support                                                  | Primary citation |
|--------------|----------------------------------------------------------------------|------------------|
| ROUGE-1/2/L  | Reference overlap over serialized answer sets and query strings      | [8]              |
| BLEU         | Sentence-level reference overlap with brevity penalty                | [11]             |
| METEOR       | Exact-token METEOR-style precision/recall with fragmentation penalty | [1]              |
| BERTScore    | Optional contextual-embedding similarity when installed              | [17]             |
| FrugalScore  | Optional efficient learned NLG metric when installed                 | [4]              |
| Cosine       | Token-count vector cosine similarity                                 | [9]              |
| Jaro-Winkler | Character-level string similarity for near-match diagnostics         | [3, 16]          |
| Exact match  | Raw and normalized string exact match                                | [14]             |

Table 20: Supplementary reference-based text metrics supported by the PIPE-Cypher evaluator. These metrics are useful for debugging answer rendering, paraphrase sensitivity, and near-match behavior, but they do not replace execution accuracy or answer-set F1 for Text2Cypher correctness. BERTScore and FrugalScore are optional integrations because they require additional metric/model packages.

*Contract.* Inputs include question, Cypher, schema slice, execution rows, and validation summary; Strict JSON scores for ambiguity, semantic alignment, schema use, and difficulty; Categorical values constrain query literals, not observed result-row values; Pass only useful, unambiguous enterprise benchmark examples

6. **Downstream Text2Cypher.** Stage: Model evaluation. SHA-256: 4c07ff.

*Contract.* Read-only Cypher only; Schema-visible constructs and exact direction preservation; RETURN DISTINCT, count/ranking rules, and no explanations

## D Representative Accepted Examples

The examples below are selected in stable identifier order from the tracked full-export snapshot, one per graph/category cell when available. They show the natural-language question, accepted Cypher, structural tags, gate status, and a bounded execution-result sample.

1. **FinBench / Boolean Existence / easy.** *Question:* Does account '187743809466009406' have any outgoing transfer?

```
MATCH (src:Account {accountId:
'187743809466009406'}) -[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT COUNT(DISTINCT src) > 0
AS HasOutgoingTransfer
```

*Structure:* single.hop, aggregation.

*Relationships:* TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {HasOutgoingTransfer: True}; observed rows: 1

2. **FinBench / Complex Aggregation / medium.** *Question:* What is the total transferred amount from accounts owned by person 'Gwar'?

```
MATCH (p:Person {personName: 'Gwar'})
-[:OWN_ACCOUNT]->
(src:Account)-[:TRANSFER_TO]->
(:Account)
```



```

RETURN DISTINCT SUM(t.amount) AS
TotalTransferredAmount

```

*Structure:* join\_heavy, aggregation.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {TotalTransferredAmount: 14972318.41}; observed rows: 1

3. **FinBench / Complex Retrieval / easy.** *Question:* Which accounts received transfers from accounts owned by person 'Zof'?

```

MATCH (p:Person {personName: 'Zof'})
-[:OWN_ACCOUNT]-> (src:Account)
-[:TRANSFER_TO]-> (dst:Account)
RETURN DISTINCT dst.accountId AS
AccountId, dst.accountType AS
AccountType, dst.isBlocked AS IsBlocked
LIMIT 300

```

*Structure:* join\_heavy, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4687402787162554854, AccountType: credit card, IsBlocked: False}; observed rows: 3

4. **FinBench / Negation Difference / medium.** *Question:* Which accounts owned by person 'Kant' have not sent any transfers?

```

MATCH (p:Person {personName: 'Kant'})
-[:OWN_ACCOUNT]-> (a:Account)
WHERE NOT (a) -[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT a.accountId AS
AccountId, a.accountType AS AccountType,
a.isBlocked AS IsBlocked
LIMIT 300

```

*Structure:* join\_heavy, negation, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4739757132830738341, AccountType: merchant account, IsBlocked: False}; observed rows: 1

5. **FinBench / Path Temporal / medium.** *Question:* Which accounts can receive money within two transfer hops from accounts owned by person 'Sossamon'?

```

MATCH (p:Person {personName:
'Sossamon'}) -[:OWN_ACCOUNT]->
(src:Account) -[:TRANSFER_TO*1..2]->
(dst:Account)
RETURN DISTINCT dst.accountId AS
AccountId, dst.accountType AS
AccountType, dst.isBlocked AS IsBlocked
LIMIT 300

```

*Structure:* join\_heavy, path, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4687402787162554854, AccountType: credit card, IsBlocked: False};  
observed rows: 4

6. **FinBench / Ranking Topk / medium.** *Question:* For accounts owned by person 'Barry', which account sent the highest total transfer amount?

```
MATCH (p:Person {personName: 'Barry'})
-[:OWN_ACCOUNT]->
(src:Account)-[t:TRANSFER_TO]->
(:Account)
WITH src, SUM(t.amount) AS totalAmount
RETURN DISTINCT src.accountId AS
AccountId, src.accountType AS
AccountType, src.isBlocked AS IsBlocked,
totalAmount
ORDER BY totalAmount DESC
LIMIT 1
```

*Structure:* join\_heavy, aggregation, order\_rank, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4732438783436260772, AccountType: internet account, IsBlocked: False}; observed rows: 1

7. **FinBench / Simple Aggregation / easy.** *Question:* How many accounts are owned by person 'Kaewsuktae'?

```
MATCH (p:Person {personName:
'Kaewsuktae'}) -[:OWN_ACCOUNT]->
(a:Account)
RETURN DISTINCT COUNT(DISTINCT a) AS
AccountCount
```

*Structure:* single\_hop, aggregation.

*Relationships:* OWN\_ACCOUNT.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountCount: 1}; observed rows: 1

8. **FinBench / Simple Retrieval / easy.** *Question:* Which accounts are owned by person 'Barry'?

```
MATCH (p:Person {personName: 'Barry'})
-[:OWN_ACCOUNT]-> (a:Account)
RETURN DISTINCT a.accountId AS
AccountId, a.accountType AS AccountType,
a.isBlocked AS IsBlocked
LIMIT 300
```

*Structure:* single\_hop, bounded\_result.

*Relationships:* OWN\_ACCOUNT.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4732438783436260772, AccountType: internet account, IsBlocked: False}; observed rows: 1

9. **SNB / Boolean Existence / medium.** *Question:* Does person with id 6597069766828 like any post?

```
MATCH (p:Person {id: 6597069766828})
OPTIONAL MATCH (p) -[:LIKES]->
(post:Post)
RETURN DISTINCT COUNT(post) > 0 AS
LikesAnyPost
```

*Structure:* single\_hop, aggregation, optional.

*Relationships:* LIKES.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {LikesAnyPost: True}; observed rows: 1

10. **SNB / Complex Aggregation / medium.** *Question:* How many distinct posts are in forums joined by person with id 6597069766845?

```
MATCH (forum:Forum) -[:HAS_MEMBER]->
(p:Person {id: 6597069766845})
MATCH (forum) -[:CONTAINER_OF]->
(post:Post)
RETURN DISTINCT COUNT(DISTINCT post) AS
JoinedForumPostCount
```

*Structure:* join\_heavy, aggregation.

*Relationships:* CONTAINER\_OF, HAS\_MEMBER.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {JoinedForumPostCount: 1}; observed rows: 1

11. **SNB / Complex Retrieval / easy.** *Question:* Which people are members of forums containing posts tagged 'Manuel\_Noriega'?

```
MATCH (forum:Forum) -[:HAS_MEMBER]->
(p:Person), (forum) -[:CONTAINER_OF]->
(post:Post) -[:HAS_TAG]-> (tag:Tag
{name: 'Manuel_Noriega'})
RETURN DISTINCT p.id AS PersonId
LIMIT 200
```

*Structure:* join\_heavy, bounded\_result.

*Relationships:* CONTAINER\_OF, HAS\_MEMBER, HAS\_TAG.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {PersonId: 4398046511220}; observed rows: 19

12. **SNB / Negation Difference / medium.** *Question:* Which members of forum 'Wall of Celso Oliveira' have not liked any post?

```
MATCH (forum:Forum {title: 'Wall of
Celso Oliveira'}) -[:HAS_MEMBER]->
(p:Person)
WHERE NOT (p) -[:LIKES]-> (:Post)
RETURN DISTINCT p.id AS PersonId,
p.firstName AS FirstName, p.lastName AS
LastName
LIMIT 200
```

*Structure:* join\_heavy, negation, bounded\_result.

*Relationships:* HAS\_MEMBER, LIKES.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {FirstName: K., LastName: Sen, PersonId: 94}; observed rows: 1

13. **SNB / Path Temporal / easy.** *Question:* Which people are within two knows hops of person with id 4398046511136?

```
MATCH (src:Person {id: 4398046511136})
-[:KNOWS*1..2]-> (dst:Person)
RETURN DISTINCT dst.id AS PersonId,
dst.firstName AS FirstName, dst.lastName
AS LastName
LIMIT 200
```

*Structure:* single\_hop, path, bounded\_result.

*Relationships:* KNOWS.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {FirstName: Rafael, LastName: Fernández, PersonId: 4398046511333}; observed rows: 25

14. **SNB / Ranking Topk / medium.** *Question:* Among forums tagged 'James\_K.\_Polk', which forum has the most members?

```
MATCH (forum:Forum) -[:HAS_TAG]->
(tag:Tag {name: 'James_K._Polk'})
MATCH (forum) -[:HAS_MEMBER]->
(person:Person)
WITH forum, COUNT(DISTINCT person) AS
memberCount
RETURN DISTINCT forum.title AS
ForumTitle, memberCount
ORDER BY memberCount DESC
LIMIT 1
```

*Structure:* join\_heavy, aggregation, order\_rank, bounded\_result.

*Relationships:* HAS\_MEMBER, HAS\_TAG.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {ForumTitle: Wall of Javed Khan, memberCount: 33}; observed rows: 1

15. **SNB / Simple Aggregation / easy.** *Question:* How many posts are tagged 'Vietnam'?

```
MATCH (post:Post) -[:HAS_TAG]-> (tag:Tag
{name: 'Vietnam'})
RETURN DISTINCT COUNT(DISTINCT post) AS
PostCount
```

*Structure:* single\_hop, aggregation.

*Relationships:* HAS\_TAG.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {PostCount: 1}; observed rows: 1

16. **SNB / Simple Retrieval / easy.** *Question:* Which post IDs did person with id 4398046511124 like?

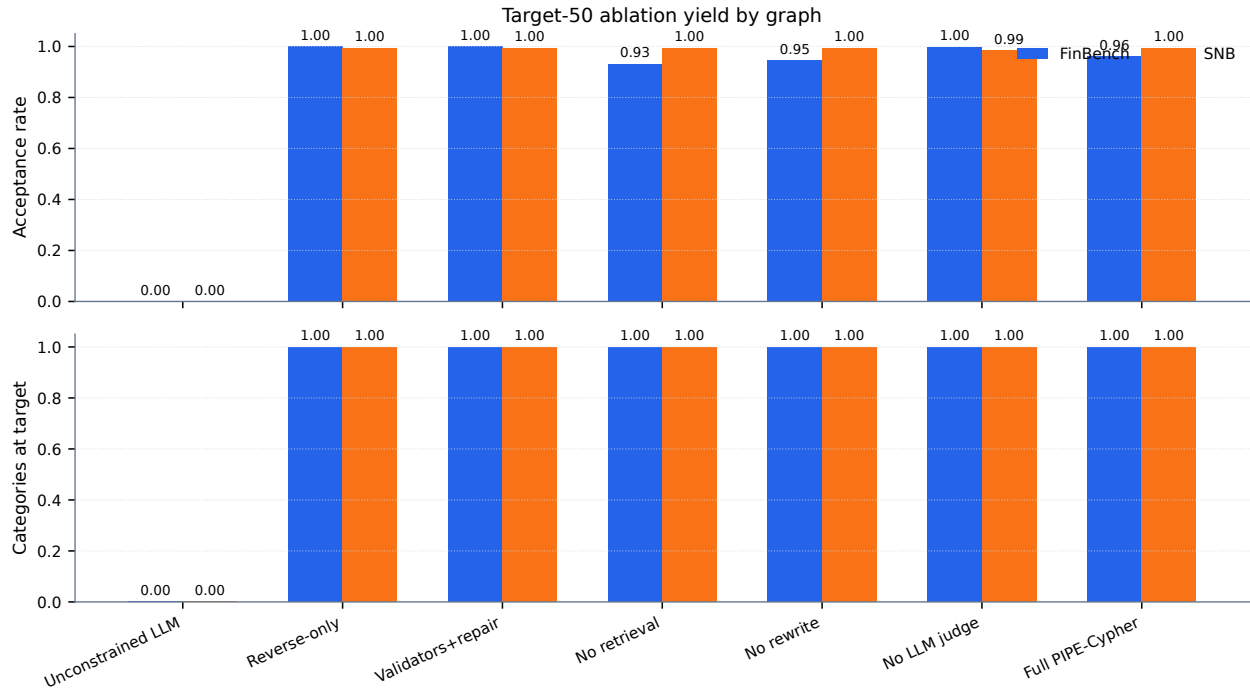


Figure 2: Audited target-50 ablation suite over FinBench and SNB. The suite reaches all graph/variant/category targets and is included as appendix evidence while the larger target-100 and seeded repeat suites continue running for stronger final reliability evidence.

```
MATCH (p:Person {id: 4398046511124})
-[:LIKES]-> (post:Post)
RETURN DISTINCT post.id AS PostId
LIMIT 200
```

*Structure:* single\_hop, bounded\_result.

*Relationships:* LIKES.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {PostId: 343597385744}; observed rows: 5

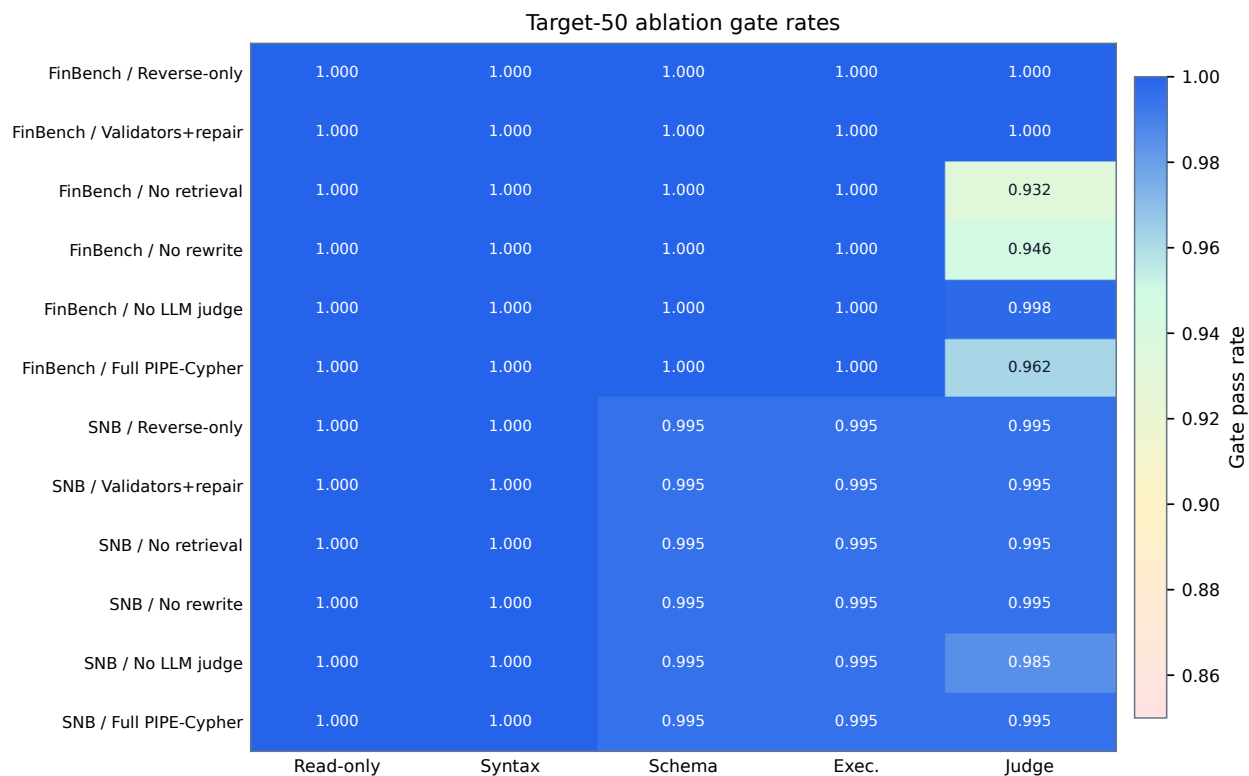


Figure 3: Gate-rate heatmap for the audited target-50 ablation suite. Deterministic read-only and syntax gates are saturated, while schema, execution, and judge rates expose the remaining quality differences across graph and pipeline variants.

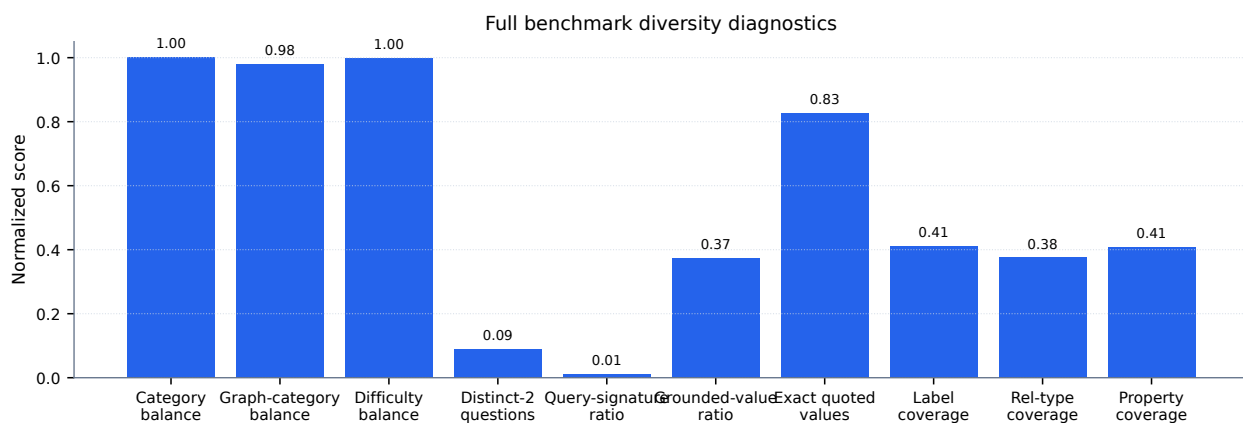


Figure 4: Diversity diagnostics for the full 3,000-example export. Category, graph-category, and difficulty balance are strong by construction; schema coverage and query-signature diversity expose remaining concentration from the seeded-template run.

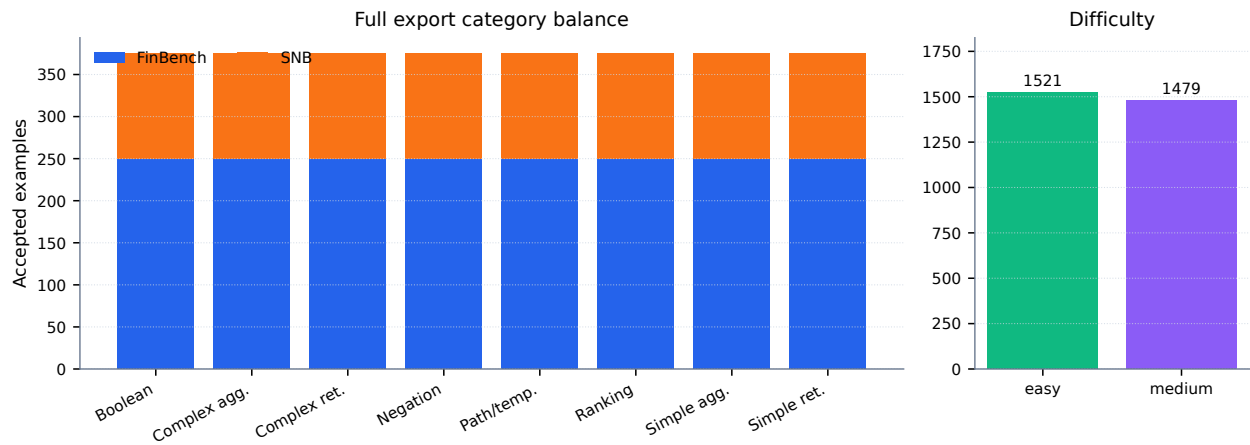


Figure 5: Full 3,000-example export distribution. The benchmark preserves the planned 2:1 FinBench/SNB graph mix while balancing all eight Cypher workload categories and maintaining a near-even easy/medium difficulty split.

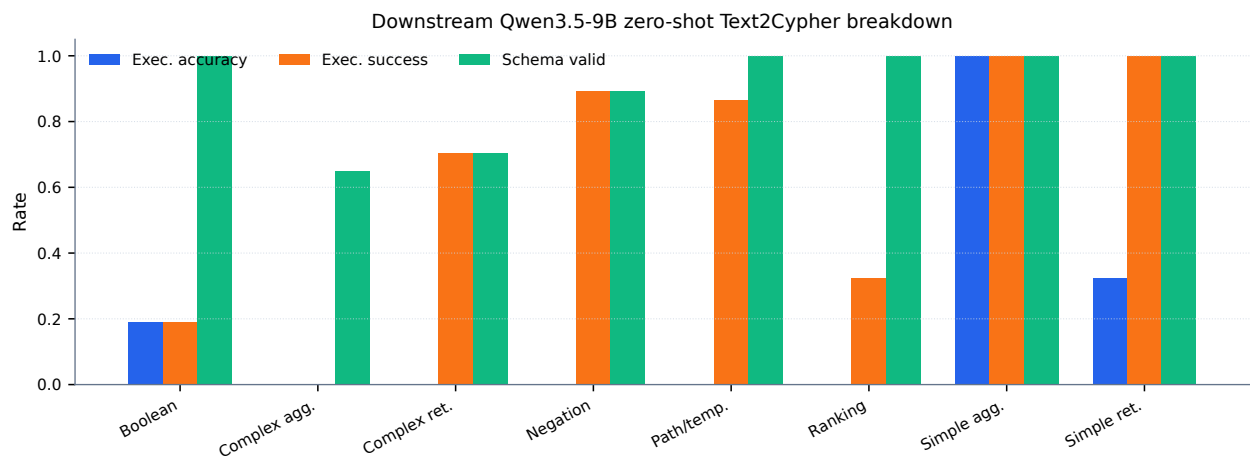


Figure 6: Downstream zero-shot Qwen3.5-9B Text2Cypher evaluation by workload category on the full test split. The breakdown separates final execution accuracy from parse/schema/execution success signals, making difficult operational categories visible.

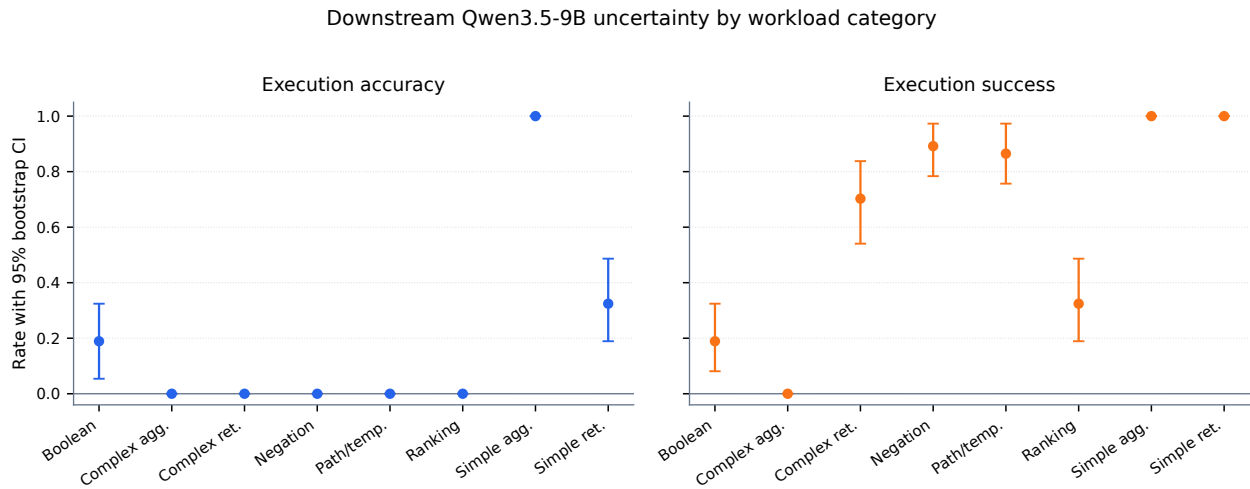


Figure 7: Bootstrap uncertainty for downstream Qwen3.5-9B Text2Cypher evaluation by workload category. Error bars show 95% row-level bootstrap intervals over the full exported test split.

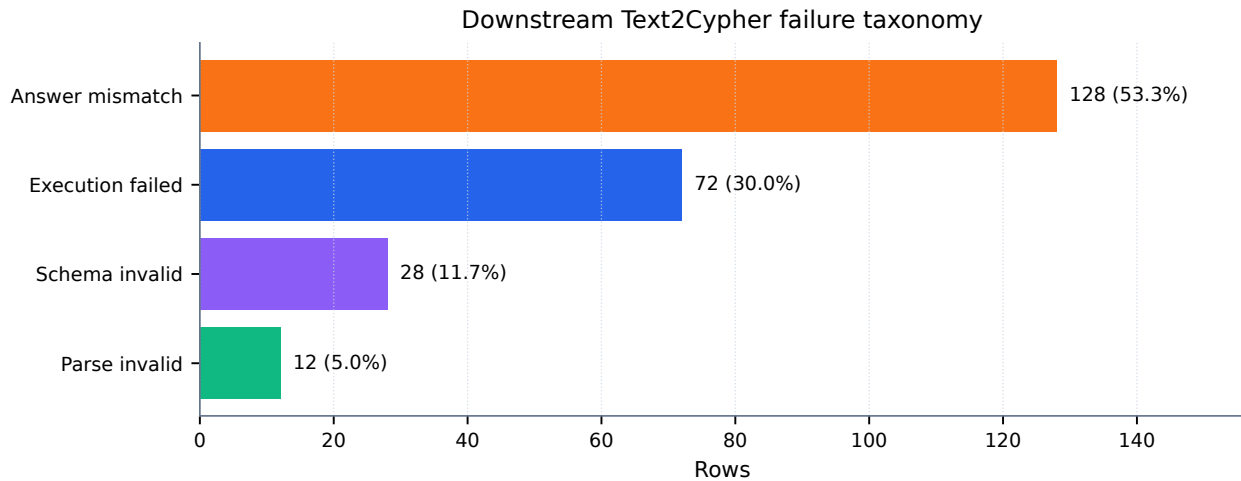


Figure 8: Downstream Text2Cypher failure taxonomy for the local Qwen3.5-9B baseline on the full test split. Exact matches are excluded so the chart exposes whether misses come from invalid Cypher, failed execution, or semantically wrong executable queries.



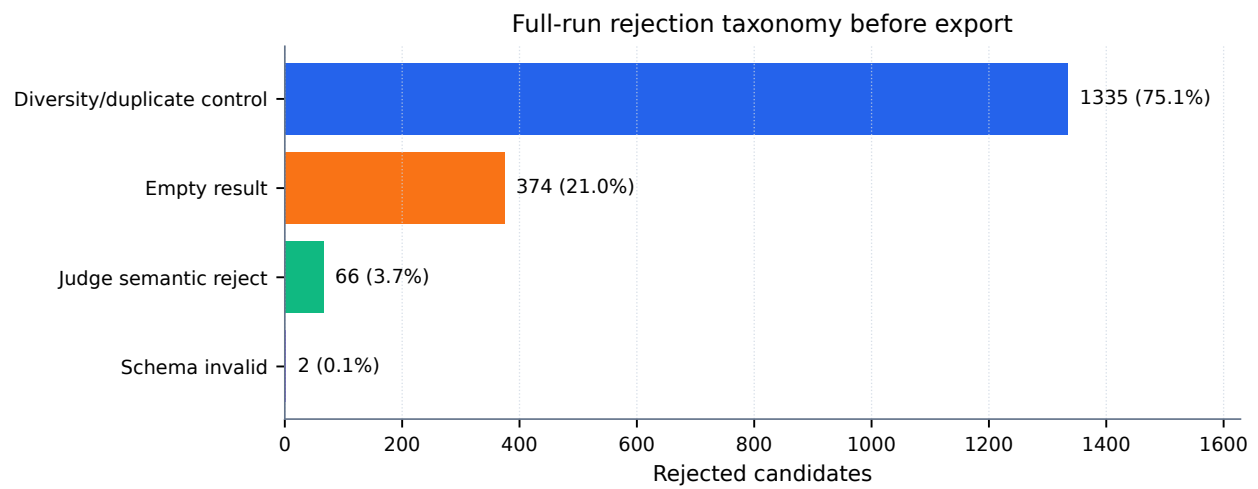


Figure 9: Full-run rejection taxonomy before benchmark export. Most rejected candidates come from duplicate/diversity control during category recovery and from empty execution results; schema-invalid Cypher is rare after deterministic governance.